



## Embedded Linux system development training

Course duration \_\_\_\_\_

 **5 days – 40 hours**

Language \_\_\_\_\_

Materials      English

Oral Lecture    English  
                         French

Trainer \_\_\_\_\_

One of the following engineers

- Cédric Madianga

Contact \_\_\_\_\_

 cedric.madianga@gmail.com

### Audience

People developing devices using the Linux kernel  
People supporting embedded Linux system developers.

### Training objectives

- Be able to understand the overall architecture of Embedded Linux systems.
- Be able to choose, build, setup and use a cross-compilation toolchain.
- Be able to understand the booting sequence of an embedded Linux system, and to set up and use the U-Boot bootloader.
- Be able to select a Linux kernel version, to configure, build and install the Linux kernel on an embedded system.
- Be able to create from scratch a Linux root filesystem, including all its elements: directories, applications, configuration files, libraries.
- Be able to choose and setup the main Linux filesystems for block and flash storage devices, and understand their main characteristics.
- Be able to interact with hardware devices, configure the kernel with appropriate drivers and extend the *Device Tree*
- Be able to select, cross-compile and integrate open-source software components (libraries, applications) in an Embedded Linux system, and to handle license compliance.
- Be able to setup and use an embedded Linux build system, to build a complete system for an embedded platform.
- Be able to develop and debug applications on an embedded Linux system.

### Prerequisites

- **Knowledge and practice of UNIX or GNU/Linux commands:** participants must be familiar with the Linux command line. Participants lacking experience on this topic should get trained by themselves, for example with our [freely available on-line slides](#).
- **Minimal English language level: B1**, according to the *Common European Framework of References for Languages*, for our sessions in English. See the [CEFR grid](#) for self-evaluation.

### Pedagogics

- Lectures delivered by the trainer: 50% of the duration
- Practical labs done by participants: 50% of the duration
- Electronic copies of presentations, lab instructions and data files. They are freely available [here](#).

### Certificate

Only the participants who have attended all training sessions, and who have scored over 50% of correct answers at the final evaluation will receive a training certificate from Bootlin.

### Disabilities

Participants with disabilities who have special needs are invited to contact us to discuss adaptations to the training course.



## Required equipment

For on-site session delivered at our customer location, our customer must provide:

- Video projector
- One PC computer on each desk (for one or two persons) with at least 16 GB of RAM, and Ubuntu Linux 26.04 installed in a free partition of at least 30 GB
- Distributions other than Ubuntu Linux 26.04 are not supported, and using Linux in a virtual machine is not supported.
- Unfiltered and fast connection to Internet: at least 50 Mbit/s of download bandwidth, and no filtering of web sites or protocols.
- PC computers with valuable data must be backed up before being used in our sessions.

For on-site sessions organized at Bootlin premises, Bootlin provides all the necessary equipment.

## Hardware platform for practical labs

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### STM32MP1 Discovery Kit

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One of these Discovery Kits from STMicroelectronics:

**STM32MP157A-DK1,**  
**STM32MP157D-DK1, STM32MP157C-**  
**DK2 or STM32MP157F-DK2**

- STM32MP157, dual Cortex-A7 processor from STMicroelectronics
- USB powered
- 512 MB DDR3L RAM
- Gigabit Ethernet port
- 4 USB 2.0 host ports
- 1 USB-C OTG port
- 1 Micro SD slot
- On-board ST-LINK/V2-1 debugger
- Arduino compatible headers
- Audio codec, buttons, LEDs
- LCD touchscreen (DK2 kits only)



### Day 1 - Morning

|         |                                         |                                                                                                                                                                                                                                                                                                                                                                   |
|---------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lecture | Introduction to embedded Linux          | <ul style="list-style-type: none"> <li>Advantages of Linux versus traditional embedded operating systems.</li> <li>Typical hardware platforms used to run embedded Linux systems.</li> <li>Overall architecture of embedded Linux systems: overview of the major software components.</li> <li>Development environment for Embedded Linux development.</li> </ul> |
| Lecture | Cross-compiling toolchain and C library | <ul style="list-style-type: none"> <li>What's inside a cross-compiling toolchain</li> <li>Choosing the target C library</li> <li>What's inside the C library</li> <li>Ready to use cross-compiling toolchains</li> <li>Building a cross-compiling toolchain with automated tools.</li> </ul>                                                                      |
| Lab     | Cross compiling toolchain               | <ul style="list-style-type: none"> <li>Getting and configuring Crosstool-NG</li> <li>Executing it to build a custom cross-compilation toolchain</li> <li>Exploring the contents of the toolchain</li> </ul>                                                                                                                                                       |

### Day 1 - Afternoon

|         |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lecture | Boot process, firmware, bootloaders | <ul style="list-style-type: none"> <li>Booting process of embedded platforms, focus on the <i>x86</i> and <i>ARM</i> architectures</li> <li>Boot process and bootloaders on <i>x86</i> platforms (legacy and UEFI)</li> <li>Boot process on ARM platforms: ROM code, bootloaders, <i>ARM Trusted Firmware</i>, <i>OP-TEE OS</i></li> <li>Focus on U-Boot: configuration, installation, and usage.</li> <li>U-Boot commands, U-Boot environment, U-Boot scripts, U-Boot generic distro boot mechanism</li> </ul> |
| Lab     | Bootloader and U-boot               | <ul style="list-style-type: none"> <li>Set up serial communication with the board.</li> <li>Configure, compile and install U-Boot, OP-TEE OS and Trusted Firmware-A for the target hardware.</li> <li>Become familiar with U-Boot environment and commands.</li> <li>Set up TFTP communication with the board. Use TFTP U-Boot commands.</li> </ul>                                                                                                                                                             |

### Day 2 - Morning

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|---------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lecture | Linux kernel                                        | <ul style="list-style-type: none"> <li>Role and general architecture of the Linux kernel</li> <li>Separation between kernel and user-space, and interfaces between user-space and the Linux kernel</li> <li>Understanding Linux kernel versions: choosing between vendor-provided kernel and upstream kernel, <i>Long Term Support</i> versions</li> <li>Getting the Linux kernel source code</li> </ul> |
| Lab     | Fetching Linux kernel sources                       | <ul style="list-style-type: none"> <li>Clone the mainline Linux tree</li> <li>Accessing stable releases</li> </ul>                                                                                                                                                                                                                                                                                       |
| Lecture | Configuring, compiling and booting the Linux kernel | <ul style="list-style-type: none"> <li>Configuring the Linux kernel: ready-made configuration files, configuration interfaces</li> <li>Concept of <i>Device Tree</i></li> <li>Cross-compiling the Linux kernel</li> <li>Study of the generated files and their role</li> <li>Installing and booting the Linux kernel</li> <li>The Linux kernel command line</li> </ul>                                   |

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| Lab | Kernel cross-compiling and booting | <ul style="list-style-type: none"> <li>▪ Configuring the Linux kernel and cross-compiling it for the embedded hardware platform.</li> <li>▪ Downloading your kernel on the board through U-boot's TFTP client.</li> <li>▪ Booting your kernel.</li> <li>▪ Automating the kernel boot process with U-Boot scripts.</li> </ul> |
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## Day 2 - Afternoon

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|---------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lecture | Root filesystem in Linux                             | <ul style="list-style-type: none"> <li>▪ Filesystems in Linux.</li> <li>▪ Role and organization of the root filesystem.</li> <li>▪ Location of the root filesystem: on storage, in memory, from the network.</li> <li>▪ Device files, virtual filesystems.</li> <li>▪ Contents of a typical root filesystem.</li> </ul>                                                                                                                                                                                                                                                         |
| Lecture | BusyBox                                              | <ul style="list-style-type: none"> <li>▪ Detailed overview. Detailed features.</li> <li>▪ Configuration, compiling and deploying.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Lab     | Tiny root filesystem built from scratch with BusyBox | <ul style="list-style-type: none"> <li>▪ Setting up a kernel to boot your system on a workstation directory exported by NFS</li> <li>▪ Passing kernel command line parameters to boot on NFS</li> <li>▪ Creating the full root filesystem from scratch. Populating it with BusyBox based utilities.</li> <li>▪ System startup using BusyBox <code>init</code></li> <li>▪ Using the BusyBox HTTP server.</li> <li>▪ Controlling the target from a web browser on the PC host.</li> <li>▪ Setting up shared libraries on the target and compiling a sample executable.</li> </ul> |

## Day 3 - Morning

|         |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lecture | Accessing hardware devices | <ul style="list-style-type: none"> <li>▪ How to access hardware on popular busses: USB, SPI, I2C, PCI</li> <li>▪ Usage of kernel drivers and direct user-space access</li> <li>▪ The <i>Device Tree</i> syntax, and how to use it to describe additional devices and pin-muxing</li> <li>▪ Finding Linux kernel drivers for specific hardware devices</li> <li>▪ Using kernel modules</li> <li>▪ Hardware access using <code>/dev</code> and <code>/sys</code></li> <li>▪ User-space interfaces for the most common hardware devices: storage, network, GPIO, LEDs, audio, graphics, video</li> </ul> |
| Lab     | Accessing hardware devices | <ul style="list-style-type: none"> <li>▪ Exploring the contents of <code>/dev</code> and <code>/sys</code> and the devices available on the embedded hardware platform.</li> <li>▪ Using GPIOs and LEDs.</li> <li>▪ Modifying the Device Tree to control pin multiplexing and to declare an I2C-connected sensor.</li> <li>▪ Adding support for an I2C-connected sensor using Linux kernel modules</li> <li>▪ Adding support for the USB WiFi adapter through an out-of-tree module.</li> </ul>                                                                                                       |

## Day 3 - Afternoon

|         |                   |                                                                                                                                                                                                                                                                                                                                                              |
|---------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lecture | Block filesystems | <ul style="list-style-type: none"> <li>▪ Accessing and partitioning block devices.</li> <li>▪ Filesystems for block devices.</li> <li>▪ Usefulness of journaled filesystems.</li> <li>▪ Read-only block filesystems.</li> <li>▪ RAM filesystems.</li> <li>▪ How to create each of these filesystems.</li> <li>▪ Suggestions for embedded systems.</li> </ul> |
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| Lab     | Block filesystems | <ul style="list-style-type: none"> <li>▪ Creating partitions on your SD card</li> <li>▪ Booting a system with a mix of filesystems: <i>SquashFS</i> for the root filesystem, <i>ext4</i> for system data, and <i>tmpfs</i> for temporary system files.</li> </ul>                                                                                |
| Lecture | Flash filesystems | <ul style="list-style-type: none"> <li>▪ The Memory Technology Devices (MTD) filesystem.</li> <li>▪ Filesystems for MTD storage: JFFS2, Yaffs2, UBIFS.</li> <li>▪ Kernel configuration options</li> <li>▪ MTD storage partitions.</li> <li>▪ Focus on today's best solution, UBI and UBIFS: preparing, flashing and using UBI images.</li> </ul> |

*Note: as the embedded hardware platform used for the labs does not have any flash-based storage, this lecture will not be illustrated with a corresponding practical lab.*

## Day 4 - Morning

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| Lecture | Cross-compiling user-space libraries and applications | <ul style="list-style-type: none"> <li>▪ Configuring, cross-compiling and installing applications and libraries.</li> <li>▪ Concept of build system, and overview of a few common build systems used by open-source projects: Makefile, <i>autotools</i>, <i>CMake</i>, <i>meson</i></li> <li>▪ Overview of the common issues encountered when cross-compiling.</li> </ul>                       |
| Lab     | Cross-compiling applications and libraries            | <ul style="list-style-type: none"> <li>▪ Manual cross-compilation of several open-source libraries and applications for an embedded platform.</li> <li>▪ Learning about common pitfalls and issues, and their solutions.</li> <li>▪ This includes compiling <i>usb-modeswitch</i> package, switch the AX900 USB Wi-Fi adapter in Wi-Fi mode and setup a wlan interface on the target.</li> </ul> |

## Day 4 - Afternoon

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| Lecture | Embedded system building tools      | <ul style="list-style-type: none"> <li>▪ Approaches for building embedded Linux systems: build systems and binary distributions</li> <li>▪ Principle of <i>build systems</i>, overview of Yocto Project/OpenEmbedded and Buildroot.</li> <li>▪ Principle of <i>binary distributions</i> and useful tools, focus on Debian/Ubuntu</li> <li>▪ Specialized software frameworks/distributions: Tizen, AGL, Android</li> </ul> |
| Lab     | System build with Buildroot         | <ul style="list-style-type: none"> <li>▪ Using Buildroot to rebuild the same basic system plus a Wi-Fi device configuration tool (<i>ex</i>) and a Wi-Fi network connection tool (<i>wpa-suplicant</i>).</li> <li>▪ Driving Wi-Fi network configuration and connection, directly from the target.</li> <li>▪ Analyzing dependencies between packages.</li> </ul>                                                          |
| Lecture | Open source licenses and compliance | <ul style="list-style-type: none"> <li>▪ Presentation of the most important open-source licenses: GPL, LGPL, MIT, BSD, Apache, etc.</li> <li>▪ Concept of <i>copyleft</i> licenses</li> <li>▪ Differences between (L)GPL version 2 and 3</li> <li>▪ Compliance with open-source licenses: best practices</li> </ul>                                                                                                       |

## Day 5 - Morning

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| Lecture | Overview of major embedded Linux software stacks | <ul style="list-style-type: none"> <li>▪ <i>systemd</i> as an <i>init</i> system</li> <li>▪ Hardware management with <i>udev</i></li> <li>▪ Inter-process communication with <i>D-Bus</i></li> <li>▪ The graphics software stack: DRM/KMS, X.org, Wayland, Qt, Gtk, OpenGL</li> <li>▪ The multimedia software stack: Video4Linux, GStreamer, Pulseaudio, Pipewire</li> </ul> |
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| Lab                      | Integration of additional software stacks | <ul style="list-style-type: none"> <li>Integration of <i>systemd</i> as an init system</li> <li>Use <i>udev</i> built in <i>systemd</i> for automatic module loading</li> <li>Use <i>systemd</i> service for automatic Wi-Fi connection at boot</li> </ul>                                                                                                                                                                                               |
| <b>Day 5 - Afternoon</b> |                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Lecture                  | Application development and debugging     | <ul style="list-style-type: none"> <li>Programming languages and libraries available.</li> <li>Build system for your application, an overview of <i>CMake</i> and <i>meson</i></li> <li>The <i>gdb</i> debugger: remote debugging with <i>gdbserver</i>, post-mortem debugging with <i>core</i> files</li> <li>Performance analysis, tracing and profiling tools, memory checkers: <i>strace</i>, <i>ltrace</i>, <i>perf</i>, <i>valgrind</i></li> </ul> |
| Lab                      | Application development and debugging     | <ul style="list-style-type: none"> <li>Creating an application that reports room temperature and humidity from an I2C-connected sensor.</li> <li>Using <i>strace</i>, <i>ltrace</i>, <i>gdbserver</i> and <i>perf</i> to debug/investigate buggy applications on the embedded board.</li> <li>Reporting live room temperature and humidity from a web interface.</li> </ul>                                                                              |
| Lecture                  | Useful resources                          | <ul style="list-style-type: none"> <li>Books about embedded Linux and system programming</li> <li>Useful online resources</li> <li>International conferences</li> </ul>                                                                                                                                                                                                                                                                                  |